

Red Blood Cells, Haemoglobin and Platelets

Hemoglobin, number and size of red blood cells, hemoglobinopathies (thalassemia, HbS, HbC), G6PD, fetal hemoglobin, reticulocytes and platelets (number, volume and function)

Sérgio F · 02-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Sérgio F does not have any of the hemoglobinopathies tested: without HbS, without HbC and without the most common beta-thalassemia variants 4 (codon 39, IVS-I-110, IVS-I-1 and IVS-I-6). It also does not have the variants of G6PD A- deficiency (V68M) or S188F (it is a man, a single copy of X, and both are normal). It does not have **JAK2 46/1** or **SH2B3 R262W** (C/C), which is good for clonal hematopoiesis. The genetic RDW is at percentile 3.9 (loci load at 3.6); a low RDW is associated with less mortality and slower biological aging. It does not have HFE C282Y. In PEAR1 it has 1 allele of lower platelet aggregation, and in ITGB3 it has PIA1/PIA1.

Unfavourable/to be considered. The genetic pattern is less red blood cells, but higher: hemoglobin at percentile 11.7 (other score at 28.0), hematocrit at 13.3, number of red blood cells at 9.5 and reticulocytes at 19.0. The mean corpuscular volume is in the 79.8-percentile 82.9 and the mean globular hemoglobin in the 74.4. The loci loads confirm the pattern: number of red blood cells in the 6.0 and VGM percentiles in the 95.4 (the hemoglobin is typical, in the 47.7, and anemia also in the 76.6). It also has the HFE H63D heterozygous, which has a slight effect.

What does it mean? The values of the hemogram of Sérgio F should tend to the lower half in hemoglobin and hematocrit and to the upper half in VGM, without this being disease. Therefore, a "normal-low" hemoglobin or a "normal-high" VGM can be genetic. Even so, they should be read alongside B12, folate and ferritin (target < 100), and never do not need to investigate a new descent. Platelets are in the middle of the distribution in the scores: the number is at the 72.4 and 41.9 percentiles and the volume in the 40.6 and 50.7. The loci load points to a slightly more numerous platelets (percentile 93.0) and smaller (MPV at 16.1), with no clinical significance.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 29 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 28 of the chip, 1 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 3 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 13 PGS Catalog scores calculated locally with 96,608 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 14252 loci, 11993 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In these scores, high percentile = higher value of the parameter (it is not risk), except in RDW, where high = unfavourable. Vuckovic 2020 scores only use significant genomic variants; Xu 2022 scores use tens of thousands of variants and explain more variance.

Hemoglobin and hematocrit

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Hemoglobin	PGS000168 (Vuckovic 2020, 515 variants)	424 515 (weighted coverage 83%)	11.7	LOW	without judgment
Hemoglobin	PGS000093 (Xu 2022, 25 thousand variants)	22,630 25,090 (weighted coverage 92%)	28.0	TYPICAL	without judgment
Hematocrit	PGS000167 (Vuckovic 2020, 502 variants)	414 502 (weighted coverage 84%)	13.3	LOW	without judgment

Red blood cells

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Red blood cells number	PGS000187 (Vuckovic 2020, 678 variants)	539 678 (weighted coverage 82%)	9.5	LOW	without judgment
Mean corpuscular volume (MCV)	PGS000176 (Vuckovic 2020, 685 variants)	529 685 (weighted coverage 79%)	79.8	TYPICAL	without judgment
Mean corpuscular volume (MCV)	PGS000101 (Xu 2022, 25 thousand variants)	22,554 25,001 (weighted coverage 91%)	82.9	HIGH	without judgment
Mean corpuscular haemoglobin (MCH)	PGS000174 (Vuckovic 2020, 628 variants)	479 628 (weighted coverage 79%)	74.4	TYPICAL	without judgment
Distribution amplitude of erythrocytes (RDW)	PGS000188 (Vuckovic 2020, 546 variants)	428 546 (weighted coverage 81%)	3.9	LOW	favourable
Reticulocytes	PGS000189 (Vuckovic 2020, 555 variants)	432 555 (weighted coverage 78%)	19.0	LOW	without judgment

- **Distribution amplitude of erythrocytes (RDW):** The high RDW is associated with more mortality and faster biological aging; therefore, low = favourable.

Platelets

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Number of platelets	PGS000186 (Vuckovic 2020, 739 variants)	578 739 (weighted coverage 81%)	72.4	TYPICAL	without judgment
Number of platelets	PGS000109 (Xu 2022, 27 thousand variants)	24,019 26,683 (weighted coverage 91%)	41.9	TYPICAL	without judgment
Mean platelet volume (MPV)	PGS000179 (Vuckovic 2020, 654 variants)	506 654 (weighted coverage 81%)	40.6	TYPICAL	without judgment
Mean platelet volume (MPV)	PGS000104 (Xu 2022, 26 thousand variants)	23,076 25,745 (weighted coverage 91%)	50.7	TYPICAL	without judgment

4. Key variants by topic

Variants with described effect. In hemoglobinopathies and in G6PD the judgment refers to disease or carrier. In counting loci the judgment is 'without judgment', because they describe variations within the normal.

Hemoglobinopathies (beta-globin) and G6PD

The HBB gene encodes beta-globin. HbS and HbC are structural variants; the 4 beta-thalassemia variants tested are the most common in the reference population. G6PD protects the red globule from oxidative stress; it is in the X chromosome, so a man has only one copy.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HBB	rs334	HbS	T/T	chip	A (0 copy/s)	favourable	HbS (Glu6Val; A na strand +: hemoglobin S, sickle cell anemia in homozygous	Strong
HBB	rs33930165	HbC	C/C	chip	T (0 copy/s)	favourable	HbC (Glu6Lys; T in strand +)	Strong
HBB	rs11549407	Cd39 C>T	G/G	chip	A (0 copy/s)	favourable	Codon 39 (Gln39Ter; A na strand +): beta-thalassemia β 0	Strong
HBB	rs35004220	IVS-I-110	C/C	chip	T (0 copy/s)	favourable	IVS-I-110 G>A: beta-thalassemia β +	Strong
HBB	rs33971440	IVS-I-1	C/C	chip	T (0 copy/s)	favourable	IVS-I-1 G>A: beta-thalassemia β 0	Strong
HBB	rs35724775	IVS-I-6	A/A	chip	G (0 copy/s)	favourable	IVS-I-6 T>C: beta-thalassemia β +	Strong
G6PD (X)	rs1050828	V68M (A-)	C (X)	chip	T (0 copy/s)	favourable	V68M (202G>A; T in strand +): variant A- of G6PD deficiency (hemolysis with fava beans, some antimalarials, sulfonamides, dapsone and rasburicase)	Strong
G6PD (X)	rs5030868	S188F	G (X)	chip	A (0 copy/s)	favourable	S188F (563C>T; A na strand +): severe variant of G6PD deficiency	Strong

Fetal hemoglobin

Fetal hemoglobin (HbF) is turned off in the first months of life by BCL11A and HBS1L-MYB axis. More HbF is not harmful; in healthy adults, these alleles only slightly affect red blood cell indices.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
BCL11A	rs1427407		G/G	chip	T (0 copy/s)	without judgment	Fetal hemoglobin repressor: the effect allele rises to HbF [GWAS Catalog: Fetal hemoglobin in Sickle cell disease (HbSS or HbSB0) (tra, allele G, beta 0.622 [0.51-0.73] unit decrease, p=2E-25, PMID 34283174] (the reported G allele reduces the trait; here the effect allele is the opposite, T)	Strong
BCL11A	rs11886868		C/T	chip	C (1 copy/s)	without judgment	Second marker of BCL11A (HbF) [GWAS Catalog: Fetal hemoglobin levels, T allele, beta 0.480 unit decrease, p=7E-35, PMID 18245381] (the reported T allele reduces the trait; here the effect allele is the opposite, C)	Moderate
HBS1L-MYB	rs9399137		T/C	chip	C (1 copy/s)	without judgment	Intergenic HBS1L-MYB: the effect allele rises to HbF and the number of red blood cells [GWAS Catalog: Hemoglobin levels, C allele, beta 0.420 [0.38-0.46] g/dl increase, p=1E-93, PMID 26366553]	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
HBG2 (XmnI)	rs7482144		A/G	imputation DR2=0.89	A (1 copy/s)	without judgment	XmnI -158C>T in gamma-globin promoter (A in strand +): more HbF under erythropoietic stress	Moderate

Iron and red blood cell production

Hepcidin (controlled by TMPRSS6, HFE and TFR2) decides how much iron reaches the marrow. The common variants move little in hemoglobin, but they add up.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TMPRSS6	rs855791	A736V	A/G	chip	G (1 copy/s)	without judgment	A736V: regulates hepcidin; a low allele of iron, hemoglobin and red blood cell volume [GWAS Catalog: Hemoglobin, G allele, beta 0.067 [0.064-0.07] unit increase, p=3E-373, PMID 34594039]	Strong
HFE	rs1800562	C282Y	G/G	chip	A (0 copy/s)	favourable	C282Y: absorbs more iron; rises hemoglobin and corpuscular volume	Strong
HFE	rs1799945	H63D	C/G	chip	G (1 copy/s)	unfavourable (light)	H63D: mild effect on iron and hemoglobin	Moderate
TFR2	rs7385804		C/A	chip	A (1 copy/s)	without judgment	2 transferrin receptor (iron sensor in the liver) [GWAS Catalog: mean corpuscular hemoglobin concentration (MCHC, mean, inv-n, allele C, beta 0.081 [0.075-0.086] unit decrease, p=4E-198, PMID 39024449] (the allele C reported reduces the trait; here the effect allele is the opposite, A)	Moderate
TFRC	rs9990333		C/C	chip	C (2 copy/s)	without judgment	1 transferrin receptor (iron uptake by erythroblasts) [GWAS Catalog: Iron status biomarkers (transferrin levels), allele T, beta 0.051 [0.037-0.065] unit decrease, p=2E-13, PMID 25352340] (the allele T reported reduces the trait; here the allele of effect is the opposite, C)	Weak
HK1	rs17476364		T/T	chip	C (0 copy/s)	without judgment	Hexokinase 1 of red blood cells: affects hemoglobin and HbA1c regardless of glucose [GWAS Catalog: Hemoglobin, C allele, beta 0.147 [0.14-0.15] unit increase, p=1E-368, PMID 32888494]	Moderate

Number, size and variability of red blood cells

Loci that decide how many red blood cells are made and of what size. SH2B3 and JAK2 46/1 also relate to clonal hematopoiesis and myeloproliferative neoplasms.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
CCND3	rs9349205		G/G	chip	A (0 copy/s)	without judgment	Cyclin D3: number of divisions of erythroblasts (more globules, smaller, or the opposite) [GWAS Catalog: Red blood cell count, allele A, beta 0.086 [0.082-0.09] SD unit increase, p=9E-360, PMID 32888493]	Moderate
SH2B3	rs3184504	R262W	C/C	chip	T (0 copy/s)	favourable	R262W: T allele increases EPO and TPO signaling: more hemoglobin, more platelets and more eosinophils	Strong

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
JAK2 (46/1)	rs12343867	46/1	T/T	chip	C (0 copy/s)	favourable	Haplotype 46/1 of JAK2: predisposes the mutation JAK2V617F and myeloproliferative neoplasms	Strong
ANK1	rs4737009		G/G	chip	A (0 copy/s)	without judgment	Ankyrin 1 (red globule membrane): HbA1c and corpuscular volume [GWAS Catalog: Glycated hemoglobin levels, allele A, beta 0.023 unit increase, p=8E-56, PMID 34059833]	Weak

Platelets: number and volume

Loci GWAS of platelet number and volume (all without judgment: variations within the normal).

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ARHGEF3	rs1354034		T/T	chip	T (2 copy/s)	without judgment	Regulates platelet size and number (effect allele increases mean platelet volume) [GWAS Catalog: Mean platelet volume, C allele, beta 0.228 [0.22-0.24] unit decrease, p=1E-833, PMID 27863252] (the reported C allele reduces the trait; here the effect allele is the opposite, T)	Moderate
BAK1	rs210134		A/A	chip	G (0 copy/s)	without judgment	Platelet apoptosis (number) [GWAS Catalog: Platelet count, G allele, beta 4.957 [4.18-5.73] 10 ⁹ /l increase, p=7E-36, PMID 22139419]	Moderate
JMJD1C	rs10761731		A/A	chip	A (2 copy/s)	without judgment	Platelet number locus [GWAS Catalog: Mean platelet volume, T allele, beta 0.126 [0.12-0.13] unit decrease, p=6E-659, PMID 32888494] (the reported T allele reduces the trait; here the effect allele is the opposite, A)	Weak
TPM4	rs8109288		G/G	chip	A (0 copy/s)	without judgment	Tropomyosin 4: production of platelets by megakaryocytes [GWAS Catalog: Mean platelet volume, allele A, beta 0.029 [0.021-0.037] ln(fl) increase, p=1E-11, PMID 22139419]	Weak

Platelets: function and aggregation

Platelet aggregation genes. The effects are small and only have interest if there is one day antiplatelet therapy.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PEAR1	rs12041331		G/A	chip	A (1 copy/s)	favourable	Intron 1 of PEAR1: allele A reduces platelet aggregation (less cardiovascular events under aspirin in some studies)	Moderate
GP6	rs1613662		A/A	chip	A (2 copy/s)	without judgment	Glucoprotein VI (collagen receptor): haplotype b reduces platelet response to collagen [GWAS Catalog: Platelet distribution width, allele A, beta 0.078 [0.072-0.084] unit increase, p=2E-143, PMID 32888494]	Weak
ITGB3	rs5918	PIA1/A2	T/T	chip	C (0 copy/s)	favourable	PIA2 (Leu33Pro; C): platelet integrin; PIA2 slightly increases aggregation (weak association with arterial thrombosis)	Weak

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each parameter, with each allele oriented to rise the parameter (or higher risk, in RDW and anemia), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Hemoglobin	1872 (1688)	1658 of 3376	1682	47.7	without judgment
Red blood cells number	2468 (2085)	1946 of 4170	2076	6.0	without judgment
Mean corpuscular volume	2692 (2198)	2213 of 4396	2188	95.4	without judgment
RDW	1207 (1002)	972 of 2004	997	3.6	favourable
Number of platelets	4044 (3382)	3475 of 6764	3370	93.0	without judgment
Mean platelet volume	1900 (1597)	1643 of 3194	1589	16.1	without judgment
Anemia	69 (41)	41 of 82	40	76.6	typical

6. Favourable

- **No hemoglobinopathies tested:** HBB HbS (T/T), HbC (C/C) and the most common 4 beta-thalassemia without the variant. Strong for these variants.
- **Normal G6PD in the two tested variants** (V68M/A- and S188F; hemizygous). Strong for these two variants; the chip does not cover the other ~200 rare variants.
- **JAK2 46/1 absent (T/T) and SH2B3 R262W absent (C/C):** less predisposition to myeloproliferative neoplasms and clonal hematopoiesis JAK2. **Moderate-strong.**
- **Low genetic RDW:** percentile 3.9 (RDW loci load at 3.6). High RDW is a marker of inflammation, erythropoietic stress and mortality. **Moderate.**
- **HFE C282Y absent** (G/G). **Strong.**
- **PEAR1 rs12041331 G/A:** 1 allele with lower platelet aggregation. **ITGB3 PIA1/PIA1** (T/T): without PIA2. **Weak.**

7. Unfavourable

- **Hemoglobin and genetically lower hematocrit:** hemoglobin at percentile 11.7 (25 thousand variants score at 28.0), hematocrit at 13.3 and hemoglobin loci load at 47.7. The hemoglobin loci load is typical (47.7), and anemia (76.6). It is not anemia, but it leaves less margin to go down. **Moderate.**
- **Less red blood cells:** percentile 9.5 (loci load at 6.0), with reticulocytes at 19.0. **Moderate.**
- **Larger lobes:** VGM in the percentiles 79.8 and 82.9 (loci load at 95.4), and MCH at 74.4. A high VGM can confuse the reading (remembers lack of B12 or folate) and may hide an incipient iron deficiency. **Moderate.**
- **HFE H63D heterozygous (C/G):** slightly higher iron absorption. **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Clonal hematopoiesis (CHIP)	JAK2 46/1 absent and SH2B3 R262W absent (less clonal expansion JAK2)	Favourable
Exhaustion of Stem Cells	Genetically lower erythropoiesis: red blood cells at percentile 9.5 and reticulocytes in the 19.0	Unfavourable (mild)
Chronic Inflammation	Genetic RDW at percentile 3.9 (high RDW accompanies chronic inflammation)	Favourable
Inflammaging	Low RDW: less age-related erythropoietic stress	Favourable
Oxidative stress	Normal G6PD in the two tested variants (red globule maintains reduced glutathione)	Favourable
Redox imbalance	Normal G6PD; no unstable hemoglobins (HbS, HbC)	Favourable
Vascular Aging	PEAR1 with 1 allele of lower aggregation; ITGB3 PIA1/PIA1; typical platelets	Favourable (light)
Fragility	Lower genetic hemoglobin (percentile 11.7): anemia in advanced age is a frailty factor	Unfavourable (mild)
Accumulation of Advanced Glycation Final Products (AGEs)	HbA1c also depends on the average life of red blood cells. HK1 and ANK1 without the alleles that rise to HbA1c	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Complete hemogram with indexes and reticulocytes** in the usual controls, dealing with this profile: hemoglobin and hematocrit tendentially in the lower half, VGM in the upper half, RDW low. A new decrease in hemoglobin or an increase in RDW deserves investigation, even within the normal. *Suggestion (not applied)*.
2. **B12, folate and ferritin** as pairs of VGM reading. B12 (1000 µg) and 5-MTHF (800 µg) are already in the plan. Ferritin remains < 100, without supplementary iron. *Suggestion (not applied)*.
3. **Rapamycin**: Mild anemia and platelet decline are known effects. With a lower genetic hemoglobin, it is desirable to see the blood count in sirolimus controls. *Suggestion (not applied)*.
4. **HbA1c**: as the globules are larger and the lower reticulocytes (red blood cells plus "old" on average), HbA1c can be read slightly above the real mean blood glucose. If HbA1c does not match glucose, cross with fructoamine or a continuous sensor. *Suggestion (not applied); weak evidence*.
5. **Blood donation**: is a natural way to maintain ferritin < 100. With lower genetic hemoglobin, it is important to confirm the hemoglobin before each donation and space the donations if they go down. *Suggestion (not applied)*.
6. **G6PD**: The tested variants are normal. Before rasburicase, primaquine, tafenoquine or dapsone, make the enzymatic dosage, because the chip does not cover all variants. *Suggestion (not applied)*.
7. **If one day appears VGM low** (microcytosis), also thinking about alpha-thalassemia, that the chip does not read. *Suggestion (not applied)*.
8. **Regular aerobic exercise, sleep and protein** (~140 g/day, as in the plan) sustains erythropoiesis. Dapagliflozin slightly rises the hematocrit, which is expected. *Suggestion (not applied)*.

Not recommended: iron in unproven supplement (target ferritin < 100); erythropoiesis stimulants.

10. Limitations

- A genotyping chip with imputation does not read the deletions of alpha-globin (alpha-thalassemia), most of the rare variants of beta-globin, the rare 200 variants of G6PD, the diseases of the red globule membrane (spherocytosis, elliptocytosis) or the rare enzymopathies (pyruvate kinase).
- Count scores explain a modest part of the variation of each parameter (5–30%). The rest depends on iron, vitamins, inflammation, drugs, training and sleep.
- JAK2 V617F and other mutations of clonal hematopoiesis are somatic (acquired) and are not seen in the DNA at birth; here only predisposition is read.
- The load of loci GWAS sums SNPs in linkage imbalance. It is a descriptive complement and does not replace the polygenic scores.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. The variants of the X chromosome (G6PD) come only from the chip, in hemizyosity. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs (HBB HbS, for example) the strand + of GRCh38 is used. In the loci taken from the GWAS Catalog with negative effect, the allele of effect presented is what rises the parameter.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: Vuckovic 2020 (PGS000167, PGS000168, PGS000174, PGS000176, PGS000179, PGS000186–PGS000189) and Xu 2022 (PGS000093, PGS000101, PGS000104, PGS000109). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Vuckovic 2020 and Chen 2020 (blood counts), Menzel 2007 and Uda 2008 (fetal hemoglobin), Benyamin 2009 (TMPRSS6), Gieger 2011 (platelets), Johnson 2010 (PEAR1), and the guideline CPIC of 2022 for G6PD. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Hemoglobinopathies (beta-globin) and G6PD	HBB	rs334	T/T	chip	A	0	strand + (palindromic; assumed +)	without the variant (non carrier)
Hemoglobinopathies (beta-globin) and G6PD	HBB	rs33930165	C/C	chip	T	0	strand +	without the variant (non carrier)
Hemoglobinopathies (beta-globin) and G6PD	HBB	rs11549407	G/G	chip	A	0	strand + (published alleles; absent site of VCF)	without the variant (non carrier)
Hemoglobinopathies (beta-globin) and G6PD	HBB	rs35004220	C/C	chip	T	0	strand + (published alleles; absent site of VCF)	without the variant (non carrier)
Hemoglobinopathies (beta-globin) and G6PD	HBB	rs33971440	C/C	chip	T	0	strand +	without the variant (non carrier)
Hemoglobinopathies (beta-globin) and G6PD	HBB	rs35724775	A/A	chip	G	0	strand + (published alleles; absent site of VCF)	without the variant (non carrier)
Hemoglobinopathies (beta-globin) and G6PD	G6PD (X)	rs1050828	C (X)	chip	T	0	strand +; hemizygous (chromosome X)	without the risk allele (X, hemizygous)
Hemoglobinopathies (beta-globin) and G6PD	G6PD (X)	rs5030868	G (X)	chip	A	0	strand +; hemizygous (chromosome X)	without the risk allele (X, hemizygous)
Fetal hemoglobin	BCL11A	rs1427407	G/G	chip	T	0	strand +	0 copy of the effect allele
Fetal hemoglobin	BCL11A	rs11886868	C/T	chip	C	1	strand +	1 copy of effect allele
Fetal hemoglobin	HBS1L-MYB	rs9399137	T/C	chip	C	1	strand +	1 copy of effect allele
Fetal hemoglobin	HBG2 (Xmnl)	rs7482144	A/G	imputation DR2=0.89	A	1	strand +	1 copy of effect allele
Iron and red blood cell production	TMPRSS6	rs855791	A/G	chip	G	1	strand +	1 copy of effect allele
Iron and red blood cell production	HFE	rs1800562	G/G	chip	A	0	strand +	without the risk allele
Iron and red blood cell production	HFE	rs1799945	C/G	chip	G	1	strand + (palindromic; assumed +)	1 copy of risk allele
Iron and red blood cell production	TFR2	rs7385804	C/A	chip	A	1	strand +	1 copy of effect allele
Iron and red blood cell production	TFRC	rs9990333	C/C	chip	C	2	strand +	2 copy of the effect allele
Iron and red blood cell production	HK1	rs17476364	T/T	chip	C	0	strand +	0 copy of the effect allele
Number, size and variability of red blood cells	CCND3	rs9349205	G/G	chip	A	0	strand +	0 copy of the effect allele
Number, size and variability of red blood cells	SH2B3	rs3184504	C/C	chip	T	0	strand +	without the risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Number, size and variability of red blood cells	JAK2 (46/1)	rs12343867	T/T	chip	C	0	strand +	without the risk allele
Number, size and variability of red blood cells	ANK1	rs4737009	G/G	chip	A	0	strand +	0 copy of the effect allele
Platelets: number and volume	ARHGEF3	rs1354034	T/T	chip	T	2	strand +	2 copy of the effect allele
Platelets: number and volume	BAK1	rs210134	A/A	chip	G	0	strand +	0 copy of the effect allele
Platelets: number and volume	JMJD1C	rs10761731	A/A	chip	A	2	strand + (palindromic; assumed +)	2 copy of the effect allele
Platelets: number and volume	TPM4	rs8109288	G/G	chip	A	0	strand +	0 copy of the effect allele
Platelets: function and aggregation	PEAR1	rs12041331	G/A	chip	A	1	strand +	1 copy of the protective allele
Platelets: function and aggregation	GP6	rs1613662	A/A	chip	A	2	strand +	2 copy of the effect allele
Platelets: function and aggregation	ITGB3	rs5918	T/T	chip	C	0	strand +	without the risk allele